ModelIt!

Student Activity Pack

Predicting the Next Pandemic

Predictive Modeling of Zoonotic Spillover and Pandemic Emergence

Grade 09L3

NGSS HS-LS2-6, HS-ETS1-4

Cause and Effect / Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where do you think new pandemic viruses come from — do they just appear, or is there a predictable process?**

2. **What do you think determines whether a new virus causes a local outbreak versus a global pandemic?**

3. **Draw a diagram showing the chain of events from a virus in a bat to a global pandemic — what steps have to occur?**

4. **What would an ideal pandemic early warning system look like, and why don't we have one?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Wildlife-Human Contact Rate
- Viral Mutation Rate
- Zoonotic Spillover Probability
- Surveillance Detection Speed
- Population Density
- Travel Connectivity
- Immune Naivety
- Healthcare Capacity
- Intervention Speed

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Slow Burn

Moderate contact, average mutation, moderate surveillance

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: COVID-19 Analog

High contact, high density, high travel, slow detection

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Early Warning Optimized

Fast surveillance, moderate risk factors

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Seasonal Climate Patterns — Temperature and precipitation cycles that affect wildlife breeding, migration, and habitat overlap with human populations — creating predictable seasonal windows of elevated spillover risk

Viral Recombination Events — The exchange of genetic material between different viral strains co-infecting the same host cell, which can rapidly create novel viruses with pandemic potential — especially concerning in intermediate hosts like pigs

Global Supply Chain Disruption — The cascade of economic, food security, and social consequences that follow pandemic-related lockdowns and travel restrictions — the secondary disaster that amplifies the human toll beyond direct disease mortality

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

COVID-19 killed millions. The next pandemic is already brewing in a bat cave, a wet market, or an industrial pig farm — can we predict where and when it will emerge? Use evidence from your simulation to support your answers.

1. **How does your predictive model identify the conditions that distinguish a contained outbreak from a global pandemic?**

2. **Which intervention point in your model has the greatest leverage for preventing pandemics — why is surveillance speed so critical?**

3. **Why do you think pandemic risk is concentrated in specific geographic hotspots rather than distributed randomly?**

4. **How would adding Seasonal Climate Patterns change your model's ability to predict spillover timing?**

5. **What are the ethical implications of predictive pandemic models — should governments restrict wildlife contact in areas your model identifies as high-risk?**

8 STEM Challenge: Design a Global Pandemic Early Warning System

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YOUR ENGINEERING MISSION

Design a surveillance network that detects zoonotic spillover events within 72 hours of the first human infection, anywhere on Earth.

THE SCENARIO

The World Health Organization has commissioned your team to design the next-generation pandemic early warning system. COVID-19 demonstrated that existing surveillance was too slow — the virus circulated for weeks or months before being identified. Your system must detect novel zoonotic viruses within 72 hours of the first human case, in any country, regardless of local healthcare infrastructure.

Engineering Design Thinking

1. HOW WOULD YOU DEPLOY SURVEILLANCE IN REMOTE AREAS WITH LIMITED HEALTHCARE INFRASTRUCTURE?
2. WHAT ROLE WOULD ARTIFICIAL INTELLIGENCE PLAY IN DETECTING OUTBREAK SIGNALS IN NOISY REAL-WORLD DATA?
3. HOW WOULD YOU OVERCOME THE POLITICAL BARRIERS TO RAPID INTERNATIONAL OUTBREAK REPORTING?
4. WHAT GENOMIC SEQUENCING CAPACITY WOULD YOUR SYSTEM NEED, AND WHERE WOULD YOU LOCATE IT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists and Pandemic Preparedness Specialists design and operate disease surveillance systems. They work for the WHO, CDC, national health agencies, and organizations like PREDICT and the Global Virome Project, earning $70,000–$150,000/year. Computational Epidemiologists who build predictive outbreak models earn $90,000–$180,000/year.